

VAMSHI KRISHNA GAREGA

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SUMMARY

Software Engineer with 4+ years building distributed data systems, full-stack applications, and AI/agent platforms. Ships end-to-end across Python (Django/FastAPI), React/TypeScript, Apache Airflow, Snowflake/Trino, and Kubernetes. At **Apple**, delivered **9 production AI agents and 13+ MCP (Model Context Protocol) servers** to **8 internal engineering teams**, owned large-scale data pipelines across **9 product lines**, and built the platform tooling that let a small team ship at scale. Known for **greenfield ownership, clean code, and turning ambiguous problems into shipped systems.**

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash, R, PHP

AI / ML & Agents: LLMs, Anthropic Claude SDK, Google ADK, MCP (Model Context Protocol / FastMCP), RAG, multi-agent systems, agent orchestration, prompt engineering, anomaly detection, ARIMA/SARIMA, TensorFlow, Scikit-Learn

Backend: Django, FastAPI, Node.js, Express.js, REST APIs, GraphQL, microservices, state-machine architecture

Frontend: React.js, Redux, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Docusaurus

Data Engineering: Apache Airflow, Apache Spark, Kafka, ETL/ELT pipelines, Snowflake, Trino/PrestoSQL, Apache Iceberg, PostgreSQL, MySQL, MongoDB, Hive, Hadoop, Databricks, Treasure Data, Azure Data Lake, AWS Redshift

Cloud & DevOps: AWS (S3, EC2, Redshift), Docker, Kubernetes, CI/CD (Rio, GitHub Actions, Jenkins), Grafana, Datadog, Splunk, Sentry

Auth / Platform: OAuth 2.0, OIDC, JWT, RBAC, secret management, token exchange (RFC 8693)

BI & Visualization: Tableau, Power BI, Plotly, Matplotlib, Seaborn, Pandas, NumPy, SciPy

Tools: Git, GitHub, VS Code, PyCharm, REST, JSON/YAML, PostMan

PROFESSIONAL BACKGROUND

Apple

Austin, Texas

Software Engineer - On-Site

May 2025 – Present

- Designed and operated **large-scale distributed data ingestion and processing systems** powering Apple's hardware manufacturing analytics platform across **9 product lines (Mac, iPhone, Watch, and more)**, transforming multi-LOB sensor telemetry from petabyte-scale cloud storage.
- Built **API-driven acquisition pipelines** with a **state-machine (controller/worker) architecture** to fetch, parse, and process manufacturing blobs at scale using **Python, Django, and Apple's internal parsing libraries**, running across **~350 production Kubernetes pods**.
- Designed, built, and deployed **9 production AI agents and 13+ MCP (Model Context Protocol) servers** using Python, FastAPI, Docker, and Kubernetes, leveraging **LLMs (Anthropic Claude SDK, Google ADK)** via **FloodGate** to automate anomaly detection, data transformation, and workflow orchestration for **8 internal engineering teams**.
- Engineered a **central MCP gateway** (FastMCP + OAuth 2.0 / OIDC) routing requests to 13+ specialized services, serving **505+ queries across ~250 users at a 95.6% success rate** and becoming the team's standard integration pattern.
- Authored **19 production Apache Airflow DAGs** and re-platformed the team's data pipeline foundation from **Python 3.6 / Airflow 1.10 to Python 3.12 / Airflow 2.7.3** (49-file foundation PR, 2,000+ LOC, 34 tests), modernizing the entire stack.
- Built a **real-time intelligent alerting system** integrating **Airflow DAGs, Slack bots, and agentic AI** to proactively notify engineers of anomalies and pipeline failures, delivering **79 alerts on its first production run** with SHA-256 deduplication and significantly improving response time and reliability.
- Designed and shipped **React 19 + TypeScript UI components** for Apple's internal analytics platform, covering configuration management, build-coverage insights, product-level analytics, and **automated Tableau dashboard embedding** for product leads.
- Built a **modern internal documentation site** with **React, Node.js, and Docusaurus**, auto-converting Markdown repositories into searchable HTML, improving cross-team onboarding and enabling non-technical contributors.
- Reduced a critical processing backlog by ~70%** by diagnosing pod-scaling bottlenecks and rescaling worker pools (25 to 85 workers), scoped pull requests without disrupting other teams.
- Authored reusable **agent and MCP bootstrap templates** and a shared base Docker image, **cutting new-service onboarding from days to ~3 hours** and powering 12+ downstream services.
- Onboarded the team to **Apple's Data Lakehouse (Apache Iceberg + Trino)**, with programmatic access ingesting **2.6M rows in the first 7 days**, and standardized credential access across **~30 services and 9 product lines** via a typed RBAC secret-management wrapper.
- Managed **CI/CD releases** across UI, backend, state machine, MCPs, and ML parsers using internal pipelines, Docker, and Git/GitHub, **monitored and resolved Kubernetes pod failures, scaling issues, and ingestion delays** via Grafana, Splunk, and CLI tooling to maintain high uptime.

Braintrust Partners LLC

Associate Data Engineer

Austin, Texas

July 2024 – April 2025

- Designed and operated **end-to-end ETL/ELT pipelines** on **Treasure Data (CDP)**, processing **millions** of customer records across ingestion, transformation, and activation stages to deliver clean, consistent data for a **leading Texas bank**.
- Automated data workflows in **Python and SQL** to refresh audience segments and sync them to **HubSpot**, **cutting manual campaign-prep effort by ~80%** and accelerating marketing activation.
- Optimized large-scale SQL queries**, **reducing query runtime by ~40%** and enabling faster churn analysis and customer segmentation for the analytics team.
- Engineered **real-time streaming pipelines with Apache Kafka** and processed structured/unstructured data on **Databricks (Apache Spark) and AWS S3**, improving data availability for downstream **AI/ML models**.
- Led **cloud data migration** from on-premises systems to **AWS Redshift, Azure Data Lake, and Snowflake**, building validation checks that ensured **zero data loss** and full integrity.
- Implemented **data-quality unit tests** and **Git-based CI** for production pipelines, improving reliability and making deployments repeatable.
- Integrated first-party data with **Facebook and LinkedIn APIs** for multi-channel activation and partnered with the Chief Data Architect on weekly client deliverables.

Texas State University

Graduate Research Assistant

San Marcos, Texas

Aug 2022 – May 2024

- Researched **time-series forecasting** by combining Adversarial Statistical Decision Theory (ASDT) with statistical models on **ERCOT energy-load data**, tackling real-world demand-prediction challenges.
- Built and optimized **ML forecasting models (ARIMA, SARIMA, HMM)**, improving **prediction accuracy by 20%** through **hyperparameter tuning and cross-validation**.
- Applied **AI/ML and adversarial-robustness techniques** to the **Joint All-Domain Command and Control (JADC2)** framework for the **U.S. Air Force Office of Scientific Research (AFOSR)**, strengthening secure decision-making under data-poisoning threats, **co-authored a peer-reviewed SPIE publication**.
- Built **data pipelines** and preprocessed large datasets with **Python (Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow)**, ensuring reproducible model training and evaluation.

Ernst And Young, LLP (EY)

Software Analyst

Bengaluru, India

Dec 2021 – July 2022

- Developed and maintained **enterprise Java applications** for business-critical systems, improving stability and performance through refactoring and **root-cause analysis**.
- Designed and built **RESTful APIs** to integrate legacy systems with third-party platforms, enabling seamless interoperability between legacy and modern components.
- Optimized algorithms and data-processing workflows using **data structures and time-complexity analysis**, improving execution efficiency in high-load environments.
- Automated deployment and system tasks in **Unix/Linux** with **shell scripting**, and collaborated cross-functionally in an **Agile** environment to resolve defects and improve reliability.

EDUCATION

Master's in Computer Science CGPA: 3.75/4.0

Texas State University

Aug 2022 – May 2024

San Marcos, TX

Bachelor of Technology in Computer Science (AI Specialization) CGPA: 3.76/4.0

Koneru Lakshmaiah Education Foundation

July 2018 – June 2022

Hyderabad, India

PUBLICATIONS

Title: "Command and Control with Poisoned Temporal Batch Data" Published in SPIE 12538 on June 12, 2023. DOI: 10.1117/12.2663

- Proposed frameworks for secure decision-making under data-poisoning attacks on temporal batch data in command-and-control systems, published in a peer-reviewed SPIE journal.

HONORS AND AWARDS

- United States Airforce Office of Scientific Research Student Fellowship Grant (~\$16,000)**
- Texas State University Student Government Scholarship (Academic Year 2023 – 2024)
- Texas State University Computer Science Graduate Academic Excellence Award (Academic Year 2023 – 2024)
- Best Graduate Research Poster for 2024 (TXST Center for Analytics and Data Science)